

Laboratory sessions

Laboratory assignment 5

Implementation of a radix-4 Booth multiplier With an AXI4 interface

Goals

- ❑ Create an AXI4 IP for the 16-bit x 16-bit radix-4 Booth multiplier developed in laboratory assignment 4
- ❑ Develop the software application for the HPS that will permit to manage the multiplier by means of interrupts
- ❑ Multiplicand and multiplier may be defined in the code or entered by terminal, the result should be displayed in terminal
- ❑ Mapping of registers in the AXI4 interface
 - Slv_reg0: 0xFF200000
 - Slv_reg1: 0xFF200004
 - Slv_reg2: 0xFF200008
 - Slv_reg3: 0xFF20000C

Mapping of data and signals in the AXI4 interface

- ❑ Input *data_a* of the multiplier is stored in *slv_reg0*.
- ❑ Input *data_b* of the multiplier is stored in *slv_reg1*.
- ❑ Output *result* of the multiplier will be stored in *slv_reg2*.
- ❑ Input *start* of the multiplier is stored in *slv_reg3(0)*.
- ❑ Output *busy* of the multiplier is mapped to *slv_reg3(1)*.
- ❑ Input *ack* of the multiplier is stored in *slv_reg3(2)*.
- ❑ If the optional part of laboratory assignment 4 is implemented, then the input *irq_enable* is stored in *slv_reg3(3)*.
- ❑ The output *irq* provided by the multiplier has to be propagated as an external output.

Sequence in the software application

- ❑ Enable the interrupts generated by the multiplier by writing a value 1 in bit 3 of `slv_reg3` (**if optional part implemented**).
- ❑ Verify that there is not a pending multiplication by checking bit 1 of `slv_reg3` (*busy* output of the multiplier).
- ❑ Clear any pending interrupts by writing a value 1 and then a value 0 on bit 2 of `slv_reg3` (*ack* input of the multiplier).
- ❑ Write the multiplicand on `slv_reg0`.
- ❑ Write the multiplier on `slv_reg1`.
- ❑ Start the multiplication by writing a value 1 on bit 0 of `slv_reg3` (*start* input of the multiplier).
- ❑ Wait for the multiplication to complete (flag set by the ISR). On completion, activate and deactivate *ack* signal and set the *start* signal to 0.
- ❑ Read the contents of `slv_reg2` (*result* output of the multiplier) and display it on the terminal by means of a call to `printf()`.